

Query Review 1

Business Logic

Display employees and their departments.

Issue

Incorrect JOIN condition. `employee_id` is not related to `department_id`.

Correct Query

```
SELECT e.first_name,  
       e.last_name,  
       d.department_name  
FROM employees e  
JOIN departments d  
ON e.department_id = d.department_id;
```

Reason

`employees.department_id` references `departments.department_id`, making this the correct relationship.

Query Review 2

Business Logic

Display employees and their job titles.

Issue

Incorrect JOIN condition. `employee_id` is not related to `job_id`.

Correct Query

```
SELECT e.first_name,  
       e.last_name,  
       j.job_title  
FROM employees e  
JOIN jobs j  
ON e.job_id = j.job_id;
```

Reason

`employees.job_id` references `jobs.job_id`, which correctly links employees to their jobs.

Query Review 3

Business Logic

Display employees who work in the Sales department.

Issue

Using `LEFT JOIN` is unnecessary because the `WHERE` clause removes unmatched rows, making it behave like an `INNER JOIN`.

Correct Query

```
SELECT e.first_name,  
       d.department_name  
FROM employees e  
JOIN departments d  
ON e.department_id = d.department_id  
WHERE d.department_name = 'Sales';
```

Reason

An `INNER JOIN` is simpler and accurately reflects the requirement.

Query Review 4

Business Logic

Display employees and their managers.

Issue

The self-join relationship is reversed.

Correct Query

```
SELECT e.first_name,  
       m.first_name AS manager_name  
FROM employees e  
JOIN employees m  
ON e.manager_id = m.employee_id;
```

Reason

manager_id stores the employee's manager, so it should be matched with the manager's employee_id.
